

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology ( M.Tech.)

Sem - II University Examination, December- 2012

M.Tech. (CAD/CAM)

ME – 718.01 Computational Fluid Dynamics

Date: 13.12.2012, Thursday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.

## SECTION- I

- Q1 (a) Explain the various components of Numerical Solution and errors in CFD tools 04
- (b) Explain what is post-processing and pre-processing in CFD. 03

- Q.2 (a) Find out whether the following differential equations are elliptic, parabolic or hyperbolic: 08

- a) The One dimensional heat conduction equation given by

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

- b) Laplace equation given by

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

- (b) Discuss in detail the three basic methods of problem solving in engineering. Also discuss their relative merits and demerits 06

OR

- Q.2 (a) State and derive the relation for Reynolds's Transport theorem. 08
- (b) Explain the CFL stability criteria. 06

- Q.3 (a) Explain what is meant by truncation error. 04
- (b) Explain the Explicit and Implicit Approach with its advantages and limitations. 10

OR

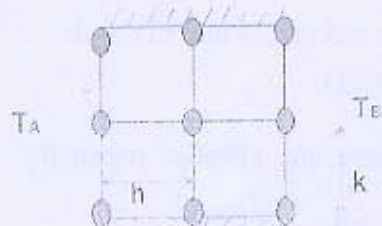
- Q.3 (a) Explain the boundary fitted co-ordinate system 03
- (b) Enlist some important area-of application of CFD. 03
- (c) Explain structured grid and Unstructured grid 08

## SECTION - II

**Q.4 (a)** List down the commercially available CFD packages available in the market. 03

**(b)** Using Taylor's series derive second order central-difference for  $\frac{\partial u}{\partial y}$ . 04

**Q.5 (a)** Calculate the temperature at described point on the plate surface, the plate of uniform thickness, homogeneous in nature, subjected to two different temperature at two ends  $T_A = 26^\circ\text{C}$  &  $T_B = 44.4522^\circ\text{C}$  and other two ends are insulated, so, insulated so involves 2D heat transfer with governing equation. 14



$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

OR

**Q.5 (a)** What are finite difference? Classify them and show their utility in solving engineering problems. Show that, in general, central finite differences are more accurate. Explain the situations where central finite difference cannot be implemented. 08

**(b)** Explain "Trapezoidal Rule" and "Simson's  $1/3^{\text{rd}}$  rule". 06

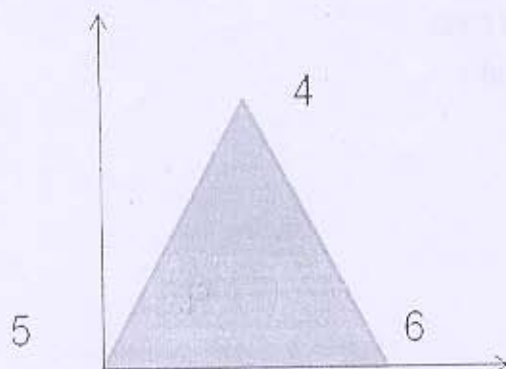
**Q.6 (a)** Explain the Various Convection Schemes (First order Upwind Scheme, Central difference Scheme, Quick Scheme). 08

**(b)** Explain physical importance of Reynold's number, weber number, and Prandtl number 06

OR

**Q.6 (a)** Discuss different types of boundary conditions with example 08

**(b)** The nodal values of all element following solution is as below. Compute the value of variable at a point P within the element as shown in fig. 06



| Node No. | X   | Y   | Variable value  |
|----------|-----|-----|-----------------|
| 4        | 3.0 | 6.0 | 154.0           |
| 5        | 0.0 | 0.0 | 454.0           |
| 6        | 6.0 | 0.0 | MM <sup>2</sup> |